Bob, verified

Hand a legacy codebase to IBM Bob and get a modernization you can trust

Team: Alan Finney, founder of 3Dogs Nexus (team of one)

3Dogs Nexus is a startup that sells consulting on problems advanced AI can solve: it builds for clients and advises clients who put their own questions to it.

IBM Bob was the engineering partner. Bob wrote every implementation module of this tool except the ported task runner, and carried out the modernization shown in this deck.

Track: Modernize What Matters

The problem

The risk is not the code Bob writes. It is trusting a step that says "done".

- AI agents change legacy code faster than a team can review it.
- Checks that always pass say nothing: a green test run on broken code is worse than no test.
- The cost ledger drifts from the real bill. In an earlier attempt by the team (15 Oct, a larger Django app, not part of this entry) the ledger recorded 15.42 credits while Bob's own task store showed 39.92.
- Afterwards nobody can see what was actually verified.

The solution

A thin layer around IBM Bob in five stages

1 Plan	Bob in plan mode: at most 3 steps, one verification command each
2 Step	Bob in agent mode implements one step at a time
3 Gate	step check + smoke checks + negative control on a deliberately broken copy; checks that cannot fail are refused (CHECKS_TOO_WEAK)
4 Ledger	per-step credits reconciled against Bob's own task store
5 Dashboard	read-only page; Re-verify now re-runs everything, no AI call, no cost, timestamped log

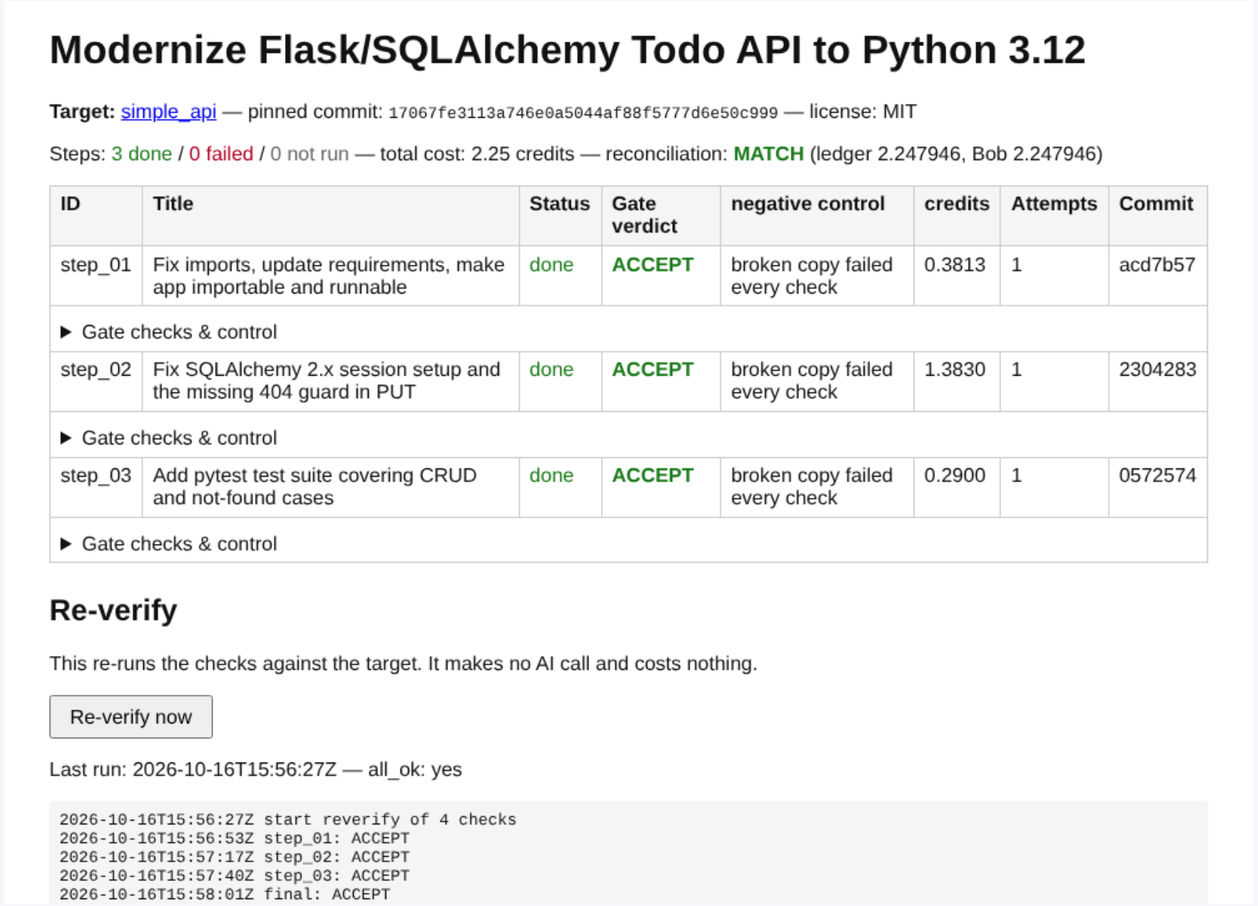
What was built, 16 to 18 Oct 2026

Python 3.12, standard library only

- Gate with negative control, ledger reconciliation, plan-to-runner wiring, dashboard with re-verify, fetch script, deploy files.
- 99 automated tests, written first by the team: including a negative test (a step that claims success but is broken is rejected) and an end-to-end test with a fake Bob.
- IBM Bob wrote the implementations in 14 small tasks for 4.22 credits; each accepted only if its tests passed and nothing else changed.
- The task runner was ported from a version the team wrote on 15 Oct, not by Bob. It is credited as such in the README and BUILD_HISTORY.

The recorded run

simple_api: a 2017 Flask todo API (MIT, 107 lines of Python) that no longer starts on current Flask



Modernize Flask/SQLAlchemy Todo API to Python 3.12

Target: simple_api — pinned commit: `17067fe3113a746e0a5044af88f5777d6e50c999` — license: MIT

Steps: 3 done / 0 failed / 0 not run — total cost: 2.25 credits — reconciliation: **MATCH** (ledger 2.247946, Bob 2.247946)

ID	Title	Status	Gate verdict	negative control	credits	Attempts	Commit
step_01	Fix imports, update requirements, make app importable and runnable	done	ACCEPT	broken copy failed every check	0.3813	1	acd7b57
▶ Gate checks & control							
step_02	Fix SQLAlchemy 2.x session setup and the missing 404 guard in PUT	done	ACCEPT	broken copy failed every check	1.3830	1	2304283
▶ Gate checks & control							
step_03	Add pytest test suite covering CRUD and not-found cases	done	ACCEPT	broken copy failed every check	0.2900	1	0572574
▶ Gate checks & control							

Re-verify

This re-runs the checks against the target. It makes no AI call and costs nothing.

Re-verify now

Last run: 2026-10-16T15:56:27Z — all_ok: yes

```
2026-10-16T15:56:27Z start reverify of 4 checks
2026-10-16T15:56:53Z step_01: ACCEPT
2026-10-16T15:57:17Z step_02: ACCEPT
2026-10-16T15:57:40Z step_03: ACCEPT
2026-10-16T15:58:01Z final: ACCEPT
```

Step	Gate / control	Credits
plan (plan mode)	3 steps	0.19
step_01 runs on 3.12	ACCEPT, 4/4 caught	0.38
step_02 SQLAlchemy 2	ACCEPT, 4/4 caught	1.38
step_03 9 pytest tests	ACCEPT, 4/4 caught	0.29
Total	**3 of 3 accepted**	**2.25**

Ledger 2.2479 against Bob's store 2.2479: MATCH. A deliberately re-broken tree was rejected (4 checks failed). Re-verify re-ran all checks against the modernized target in 94 seconds, no AI call: all ACCEPT.

Live demo: https://18-216-235-168.sslip.io/

Against the judging criteria

Tool usage and depth (30)

Bob as builder, planner and implementer: 18 tasks (14 building the tool, 1 plan mode, 3 agent mode on the target). Bob-Task and Bob-Trace trailers on every Bob commit. Cost reconciled to Bob's own store, difference 0.0000.

Innovation and use case (30)

The negative control: a check is trusted only if it fails on a broken copy. Plus cost reconciliation and zero-cost re-verification: AI-assisted modernization you can audit.

Functionality and execution (30)

99 tests green. Real end-to-end run on a real legacy app: 3 of 3 steps accepted, every control caught, rejection demonstrated, real timestamped re-verify. Live on a credential-free server; deployable with systemd or Docker.

Presentation (10)

Live dashboard, README that states the contribution plainly, 3-minute script.

Roadmap and key learnings

Roadmap

- More targets and languages; parallel steps with merge checks.
- A hosted service: upload a repository, receive a verified modernization with its evidence.
- Per-step spending limits that stop a run automatically.

Key learnings

- Small tasks with exact tests are cheap: 14 tool-building tasks for 4.22 credits, against long multi-step tasks that hit turn limits earlier.
- Reconcile the ledger against the tool's own store; do not trust your own bookkeeping.
- A check is only as good as its ability to fail.
- The human's best contribution is the contract (tests and specifications), not the code.